

Geometry
Unit 13
Lesson 10 Homework

Name Answer Key
Date _____
Period _____

Complete the table for each given image.

Question 1	Work		Answer
			Properties
	Slope	$\overline{TP}: 1, \overline{RA}: -1$ $\overline{PA}: \text{undefined}$ $\overline{TR}: \text{undefined}$	Congruent legs. Congruent diagonals. One set of parallel sides.
	Side Lengths	$RT = 6$ $TP = \sqrt{2^2 + 2^2} = 2\sqrt{2}$ $PA = 10$ $AR = \sqrt{2^2 + 2^2} = 2\sqrt{2}$	
	Diagonal Lengths	$RP = \sqrt{2^2 + 8^2} = 2\sqrt{17}$ $TA = \sqrt{2^2 + 8^2} = 2\sqrt{17}$	Quadrilateral Isosceles trapezoid
Question 2	Work		Answer
			Properties
	Slope	$\overline{KB}: \frac{3}{5}, \overline{MS}: \frac{8}{5}$ $\overline{BM}: -\frac{3}{5}, \overline{KS}: -\frac{8}{5}$	Two sets of adjacent congruent sides. No parallel sides.
	Side Lengths	$KB = \sqrt{3^2 + 5^2} = \sqrt{34}$ $BM = \sqrt{3^2 + 5^2} = \sqrt{34}$ $MS = \sqrt{8^2 + 5^2} = \sqrt{89}$ $KS = \sqrt{8^2 + 5^2} = \sqrt{89}$	
	Diagonal Lengths	$SB = 11$ $KM = 10$	Quadrilateral Kite

For questions 3 and 4, given the coordinates, classify the quadrilaterals based on their properties.

Question 3	Work		Answer
			Properties
$SKWC$ has vertices at $S(6, -3)$, $K(1, 3)$, $W(-5, -2)$, and $C(0, -8)$. $SKWC$ is classified as a(n) square because... 4 congruent sides and 4 right angles	Slope	$\overline{SK}: -\frac{6}{5}, \overline{WC}: -\frac{6}{5}$ $\overline{KW}: \frac{5}{6}, \overline{CS}: \frac{5}{6}$	Perpendicular consecutive sides (right angles). 4 congruent sides and congruent diagonals. 2 sets of parallel sides.
	Side Lengths	$SK = \sqrt{5^2 + 6^2} = \sqrt{61}$ $KW = \sqrt{5^2 + 6^2} = \sqrt{61}$ $WC = \sqrt{5^2 + 6^2} = \sqrt{61}$ $CS = \sqrt{5^2 + 6^2} = \sqrt{61}$	
	Diagonal Lengths	$SW = \sqrt{11^2 + 1^2} = \sqrt{122}$ $KC = \sqrt{11^2 + 1^2} = \sqrt{122}$	Quadrilateral Square
Question 4	Work		Answer
			Properties
$GNTH$ has vertices at $G(5, 8)$, $N(1, 1)$, $T(0, -7)$, and $H(4, 0)$. $GNTH$ is classified as a(n) rhombus because... 4 congruent sides with no right angles.	Slope	$\overline{GN}: \frac{7}{4}, \overline{TH}: \frac{7}{4}$ $\overline{NT}: \frac{8}{1}, \overline{HG}: \frac{8}{1}$	Two sets of parallel opposite sides. All 4 sides congruent. No perpendicular sides. Diagonals not congruent.
	Side Lengths	$GN = \sqrt{4^2 + 7^2} = \sqrt{65}$ $NT = \sqrt{1^2 + 8^2} = \sqrt{65}$ $TH = \sqrt{4^2 + 7^2} = \sqrt{65}$ $HG = \sqrt{1^2 + 8^2} = \sqrt{65}$	
	Diagonal Lengths	$GT = \sqrt{5^2 + 15^2} = 5\sqrt{10}$ $NH = \sqrt{3^2 + 1^2} = \sqrt{10}$	Quadrilateral Rhombus